

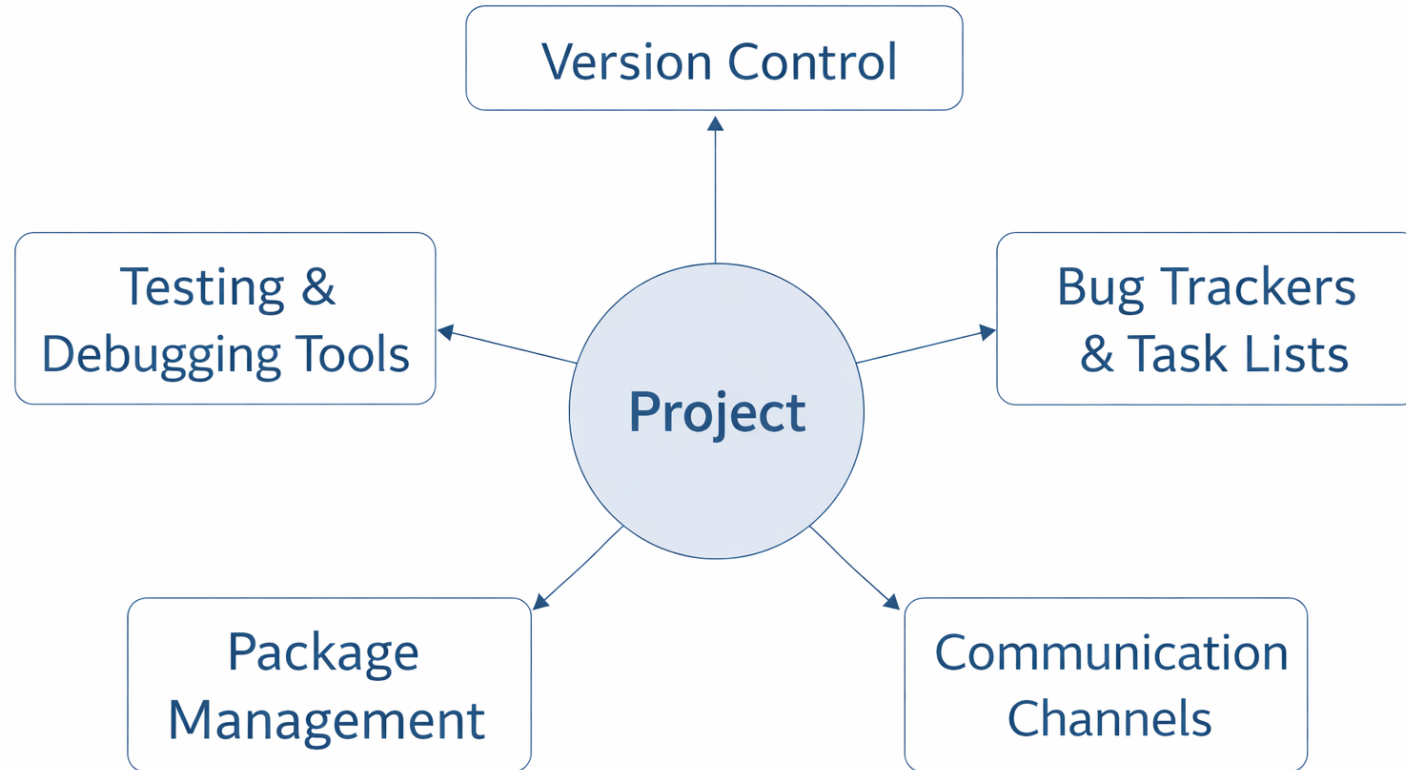


Introduction to Open Source Software

Junaid Rashid

- 
- **Open Source Tools**
 - **Starting and Managing an OSS Project**

Tools



Version control

- Keep track of file changes
- Easy to revert to previous file versions
- Option to view changes made to files
- “Source code management” or “revision control system”



git



SUBVERSION



Bug trackers and task lists

Simple and effective tools for tracking errors and failures in the development of the project.

Issues Tracker



Jira Software



GitHub
Issues

Bug trackers and task lists

Open	36	Closed	1,349	Author	Labels	Projects	Milestones	Assignees	Types	≡↓ Newest
•	There is currently no Nigerian TIV (tiv) translation available for the GitHub Desktop tutorial.	discussion	question		10					
	#114289 · Akansrodger opened 3 weeks ago									
•	Add kinyarwanda translated readme & video tutorial suggestion	discussion	question		1					
	#114104 · B-Pine opened 3 weeks ago									
•	Add Swahili (Kiswahili) Translation for GitHub Desktop Tutorial				1					
	#111788 · SRupia opened on Feb 14									
•	github cli tutorial in swahili.				2					
	#111583 · Famaskah opened on Feb 11									
•	Update content of README.sw.md	discussion	question		2					
	#111578 · Famaskah opened on Feb 11									
•	Auto-merge workflow skips comment when Contributors.md has multiple line changes	discussion	question		4					
	#111414 · Sunil0620 opened on Feb 7									
•	Link for Slack is no longer active on Home Page	discussion	question		13					
	#111088 · AnkitaPihulkar opened on Jan 31									
•	'Go to Project' button does not work on few left-side project cards	discussion	question		9					
	#110682 · CSroseX opened on Jan 23									
•	Add Montenegrin translation of README	discussion	question		7					
	#109221 · Damir1110 opened on Dec 25, 2025									
•	Add Turkish translations for CLI and GUI tool sections	discussion	question		6					
	#109155 · SMeydan opened on Dec 24, 2025									
•	Unavailable Yoruba flag				2					
	#109128 · ramma opened on Dec 24, 2025									

https://github.com/firstcontributions/first-contributions/issues?utm_source=chatgpt.com

Communications Channels

Community weekly calls

Email

Internet relay chat



What is Git?

Git is a distributed version control system that:

- Tracks changes to content
- Provides a central point for collaboration
- Enables continuously integrated changes



What is Git?

- Git enables centralized administration
 - Teams have controlled access scope
 - The main branch should always correspond to deployable code



What is GitHub?

- GitHub is an online hosting service for Git repositories.
 - Hosted by a subsidiary of Microsoft.
 - Offers free, professional and enterprise accounts.
 - Has over 190 million repositories and 40 million users.



What is GitHub?

- Collaborative coding
- Automation & CI/CD
- Security
- Client apps
- Team administration
- Project management





Starting and Managing an Open Source Project

How to Start an Open Source Project

- ❑ 1. Define project idea
- 2. Create GitHub repository
- 3. Add README file
- 4. Choose license
- 5. Add basic code/files
- 6. Make repository public

Workflow



1. Fork repository
2. Clone project
3. Create branch
4. Make changes
5. Commit & push
6. Create Pull Request
7. Maintainer reviews & merges

Releasing your own project

- ❑ README file
- ❑ Contributing guidelines
- ❑ Code of Conduct (CoC)
- ❑ Open source license

Managing an Open Source Project

- Review pull requests
- Manage issues (bugs, features)
- Accept or reject changes
- Keep code clean and updated
- Guide contributors

Roles in Open Source Project

- Maintainer → Manages project
- Contributor → Adds code/features
- User → Uses project & reports issues



□ Best Practices

- Write clear README
- Use meaningful commit messages
- Keep code simple
- Respond to issues quickly
- Be respectful to contributors

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- Today we will do:
 1. Create GitHub repository
 2. Add README file
 3. Create & edit file
 4. Commit changes
 5. Upload file
 6. Create branch
 7. Add new feature
 8. Create Pull Request
 9. Merge Pull Request

Class Task

□ Working with a GitHub repository

The steps are uploaded on the portal as a PDF file.

Step 1: In this task, you will get started with GitHub by creating a GitHub repository. You will then add a file to the repository using the GitHub web interface.

Step 2: Working with branches